The Amazon Fund and Deforestation

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1 Abstract

> "They are doing a lot of nonsense, and they are going to do more nonsense. Nature is always giving things, and men are always destroying them with fire. An unlimited donation to an eternal ingratitude. [...] It's so elementary I shouldn't even be talking here. But you have to talk, because that's what they're doing: burning the bushes and killing the lakes"[1]

This seminar paper aims to evaluate the impact of sustainable projects financed by the Amazon Fund, between 2008 and 2020, in Brazil. The projects have an economic incentive that is combined with the Fund's main goal of reducing deforestation levels in the Amazon forest. Using deforestation data at the municipal level, and exploring the time and geographic variation of the Funds resource allocation, led to non-conclusive results about the Funds impact. At the macro-level, no causal effect was found.

2 Introduction

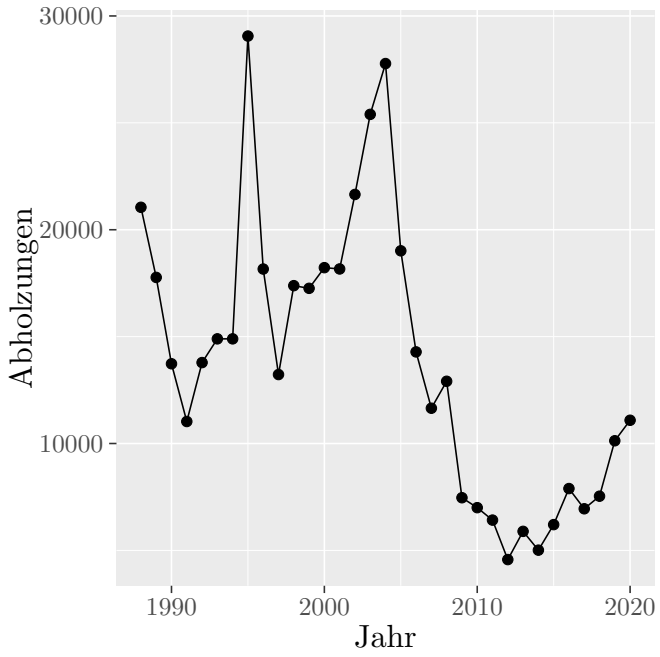


Figure 1: Cleared areas in km^2 in the Brazilian Amazon

[1] Roberto D'Avila interviews Antonio Carlos Jobim (1981)

With alarming threats presented by climate change, the discussion about forest preservation has become more urgent than ever. Deforestation is one of the main challenges of environmental policy, as it presents countless environmental and social costs like the loss of biodiversity, greenhouse gas emissions as well as the services provided by forests, such as the absorption of carbon dioxide.

The Amazon is the world's largest Forest with an area, in Brazil, of more than 4 million square kilometers - an area equivalent to almost half of Europe. Apart from its large biodiversity and water supplies, it serves as an important regulator of world climate.

Although it is in the interest of both the people living in and outside of the Amazon to preserve it, the deforestation rates have substantially increased in recent years, mainly due to a radical change in environmental policies. The topic is, on that account, extremely relevant for Brazilian environmental authorities and people living in the Amazon.

It is important to note, as shown in figure 1, that until the beginning of the 2010's Amazon deforestation rates were sharply declining. After a peak of almost 28000 square kilometers in 2004, the deforestation rates went down to about 4700 square kilometers in 2012. In 2020, however, the cleared area was about 11000 square kilometers, an area roughly equal to the size of Jamaica. Figure 2 mirrors the values and trends of figure 1 from the period 2008-2020. Instead of aggregate values, the figure is showing year values for each municipality in the Amazon. Because there are 509 municipalities on the sample, the points end up not saying much about any municipality, but the resulting convex shape is the replication of the most recent values of Figure 1.

2.1 The Fund

The goal of this seminar paper is to evaluate the aggregated contribution of projects financed by the Amazon Fund to the 2000's and 2010's decline in deforestation rates, and possibly to the rise of 2015 onwards. The Fund was created in 2008 and suspended by the federal government in 2019, thereby, extending through a period of sharp declines followed by sharp increases in deforestation levels. The Fund financed projects led by NGO's, private citizens as well as environmental authorities at the municipal, state and federal level. There are many types of projects financed by the Fund, like projects related to sustainable farming and fishing, the management of indigenous reservations, reforestation and even spatial vigilance of deforestation. The most important ones, i.e. the ones that receive more resources, nonetheless, are the ones directly pointed at environmental authorities. Until 2020, the Funds supported 102 projects with 705 million USD, being directly responsible for 1236 environmental inspection missions. About 193.000 people and 338 institutions benefited from sustainable productive activities promoted by the fund, and its territorial management extends through an area of 730.000 square kilometers, half of that area being in indigenous lands. Finally, the Fund was financed by the Brazilian oil company *Petrobras Brasileiro S.A.*, the KfW Entwicklungsbank of Germany and, almost entirely, by the Ministry of Foreign Affairs of Norway.

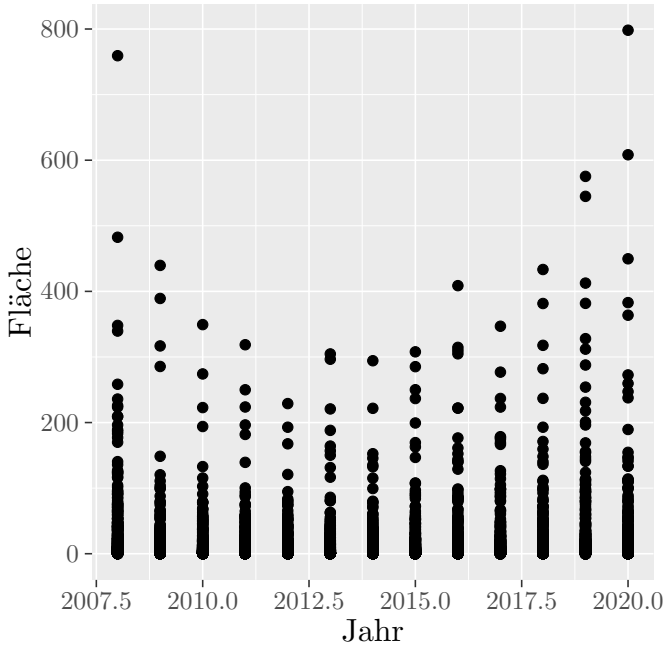


Figure 2: Yearly cleared area in km^2 for municipalities

Because the allocation of many projects at the municipal level are focused in municipalities with greater presence of higher deforestation rates, the presence of the Fund in a municipality is highly correlated with the presence of illegal deforestation activities. Finding a causal effect, therefore, depends on untwining both effects. Literature in the context of police presence and crime is vast and uses different approaches to isolate one effect from another. Assunção, J., Gandour, C., & Rocha, R., (2013) face a very similar problem dealing with law enforcement and deforestation, "Because the allocation of law enforcement teams typically targets areas under greater risk of deforestation, the correlation between the presence of the enforcers and forest clearings is jointly determined by the risk-based targeting strategy and the potentially deterrent effect of command and control"(p.3).

Literature on the impact of the Amazon fund is predominantly limited to the funds own reports and independent evaluations which can be found on their website[2]. Apart from evaluations on individual projects, an ex-post evaluation of the Fund by the KfW Entwicklungsbank can also be found in the Funds website.

This analysis uses the variation in Fund resource allocation, the Treatment, across municipalities in the so called Legal Amazon, an area that includes 772 municipalities distributed over 9 states. Combining the panel data for deforestation rates with Fund

[2]http://www.fundoamazonia.gov.br/en/monitoring-evaluation/independent-evaluations/

data serving as a treatment variable allows for several empirical approaches wich will be presented in the empirical strategy section.

3 Data and Descriptive Statistics

3.1 A. Data

The data used in this seminar comes from several sources. Panel Data for deforestation rates was found on the *terrabrasilis* web portal, a platform developed by INPE [3]. The panel data available for download includes the yearly cleared area for 640 municipalities over the 2008-2020 period. Since many municipalities have missing data for multiple years, the balanced panel used for the analysis only includes 509 municipalities, with 6617 observations in total[4].

Data for the Amazon fund was found directly on their portal. There was a total of 102 projects that summed up costed 1.8 billion R$, which were entirely financed by the fund. Out of these, only 65 were limited to municipal activities, extending through 151 different municipalities and costing about 261 million R$. These projects were often financed across multiple years, and multiple municipalities had more than one project over the years, which is a challenge for the treatment approach. The data, however, is detailed to how much money was invested in each project and municipality over the years.

Additionally, the *terrabrasilis* datasets also includes the indigenous reservation units in the Amazon. This dataset has the total area of a reservation unit as well as the dates when it was homologated and to which municipalities it belongs to. Forest losses within a reservation unit are, on average, lower than outside the borders of that unit [1], making it a relevant control to be inserted in the model.

Geographic data for municipal total areas, and areas used in agriculture were all found at the IBGE portal. This dataset also includes the areas used in the production of specific crops, which unfortunately were not used for the analysis. The prices of the main crops were used however, and they all come from the FRED portal. The prices are time-variant which allows them to be used in the panel data approach. Agriculture is an interesting control, because until the end of the 2000's it was highly correlated to deforestation [2]. An interesting effect is found for this control, which is shown in the appendix.

3.2 Descriptive Statistics

Table 1 and Table 2 are showing descriptive statistics for the sample of treated units, with 145 units and 1885 observations, and for the whole sample, respectively. The state column is showing observations from a given state, not the amount of municipalities coming from that state. Column 2 of both tables is displaying the cleared areas on both samples. The

[3] Institute for spatial research responsible for the DETER and PRODES satellite monitoring systems

[4] 13 for each municipality

comparison between both these columns already points to the conclusion that municipalities with higher deforestation activities might have been targeted or selected by the fund. It is worth pointing out, however, that although the cleared areas for treated units was, on average, more than double that of the whole sample, the average total area of treated units is also almost twice as large as the average for the whole sample.

Interestingly, the maximum value for those columns is the same for both samples. It comes from a municipality called *Altamira*, which has historically been one the municipalities with the highest annual cleared areas. The fact that this municipality received over half a million R$ from the Amazon fund in the period from 2017 to 2020, when its deforestation rates were rising again (eventually reaching the highest cleared annual area of the sample in 2020), might illustrate the way the fund acted during its existence.

Columns 3 and 5 are displaying areas used in agriculture and indigenous reservation areas respectively. Until the beginning of the 2000's commodity prices were highly correlated with deforestation rates in the Amazon. This correlation ceased to exist in the 2010's as agriculture productivity started increasing. Historically, the deforestation within a municipality that has indigenous reservation is higher outside the borders of the indigenous area. These areas are, on average, relatively higher for treated units.

State	Cleared Area	Agricultural Area	Total Area	Reservation Area
Pará :686	Min. : 0.0607	Min. : 1.0	Min. : 561.7	Min. : 0
Mato Grosso:370	1st Qu.: 2.5604	1st Qu.: 109.0	1st Qu.: 3207.8	1st Qu.: 0
Amazonas :271	Median : 10.3047	Median : 572.5	Median : 7067.1	Median : 6524
Rondônia :270	Mean : 32.1595	Mean : 5095.1	Mean : 15643.9	Mean : 357325
Acre :234	3rd Qu.: 30.4132	3rd Qu.: 2485.2	3rd Qu.: 17423.0	3rd Qu.: 189500
(Other) : 39	Max. :798.1954	Max. :121199.0	Max. :159533.3	Max. :7731691
NA's : 15		NA's :13		

Table 1: Treated municipalities - Statistics

State	Cleared Area	Agricultural Area	Total Area	Reservation Area
Pará :1531	Min. : 0.0000	Min. : 0	Min. : 127.2	Min. : 0
Mato Grosso :1216	1st Qu.: 0.8011	1st Qu.: 17	1st Qu.: 1331.7	1st Qu.: 0
Maranhão :1186	Median : 2.8767	Median : 160	Median : 3773.5	Median : 0
Amazonas : 781	Mean : 14.4027	Mean : 2458	Mean : 8828.7	Mean : 217300
Rondônia : 655	3rd Qu.: 11.2137	3rd Qu.: 873	3rd Qu.: 8783.5	3rd Qu.: 86935
(Other) :1042	Max. :798.1954	Max. :173086	Max. :159533.3	Max. :10012561
NA's : 206		NA's :156	NA's :26	

Table 2: Entire sample - Statistics

4 The Empirical Framework

To estimate the effect of projects financed by the Amazon Fund on deforestation rates over the period of 2008-2020, the following linear specification and estimate equations were used:

$$Y_{it} = \alpha + \theta Treated_{it} + \gamma Year_t + \delta TE_{it} + \epsilon \tag{1}$$

The outcome Y_{it}, is the cleared forest area on municipality i in year t, measured in square kilometers. $Treated_{it}$ is a treatment indicator which is equal to 1 for the period, as well as all subsequent periods, in which a municipality had a project financed by the Amazon fund. Once a municipality is has a project financed by the fund, it counts as a treated municipality. $Year_t$ are year dummies. The TE_{it} variable is the interaction between the $Treated_{it}$ variable and the $Year_t$ variable. δ is the main coefficient of interest measuring the treatment effect for a treated municipality in years after treatment.

The error term ϵ may be correlated with the treatment indicator variable, due to omitted variables related to environmental policies or even more generally to political variables, since the fund itself is managed by a political institution. It isn't clear, however, what the direction of the omitted variable bias is, as the political variable affecting the outcome could be positively or negatively correlated with treatment. In the context of Brazil that is not surprising, since the federal government has a large influence in resources allocated to states and municipalities. There are several other public policies aimed at the environment that could be correlated with the treatment such as *black-listing* municipalities, the rural environmental Cadaster [3] and, most likely, corruption [4], that will be omitted from this analysis.

Using the aforementioned equation, the main method, namely a staggered Difference-in-Difference is used to estimate the Funds treatment effect on deforestation rates across municipalities. The linear specification is also used for Fixed effects models (the within estimator) yielding similar results to the Difference-in-Difference estimator. Additionally, controls and clustered standard errors are also used to test the models.

4.1 Identification Strategy

The time and geographic varying allocation of the funds resources allows for a Difference-in-Difference approach. Because the treatment is continuous (projects can be financed in more than one period), and the adoption dates of treatment vary across units, a staggered Difference-in-Difference is used in this analysis.

One of the main assumptions for the staggered Difference-in-Difference approach is the Exogeneity Assumption. In practice, it means that the rollout of of projects across municipalities has to be exogenous and as good as randomly implemented. A randomized treatment ensures that it is not related to other explanatory variables affecting the outcome,

or the outcome itself. If the municipal trends are correlated to the introduction of treatment as well as the outcome, at the same time, the exogeneity assumption is not fulfiled [5].

In the case of the Amazon fund, it is likely that treatment was endogenous: the fund intervened in municipalities as a response to their increasing, or increased, deforestation rates, or for reasons that may be correlated to the outcome, like the aforementioned political omitted variable.

A staggered Difference-in-Difference also requires two exclusion restrictions [6], the first of them being the no-anticipation assumption, which implies that future treatment dates do not affect the current outcome. Anticipation effects could lead to differences in pre-treatment outcomes, also leading to disruptions in the common trend assumption. Municipalities could change their behavior, their environmental activities and policies, expecting they would be treated in the future. Although it is hard to rule out Anticipation in this case, one can expect the differences in pre-trends are likely originating from endogeneity. Municipalities can/will only expect to be treated in the future, and therefore change their behavior in anticipation, if they have access to information on future treatment, or if they benefit by acting before treatment is adopted [7]. Even if that is the case, and municipalities start changing their behavior, regarding deforestation, prior to treatment, it is still unclear how exactly the anticipation effects would influence the pre-treatment outcomes.

The most important and classic assumption for a Difference-in-Difference approach is the Common (or parallel) Trend Assumption. Intuitively it ensures that in absence of treatment, control and treatment units would've followed the same trend. This crucial Assumption is necessary for causal inference. It is also the assumption most likely not to be fulfilled in the Amazon fund case, due to reasons related to the endogenous treatment: by treating municipalities with increasing deforestation rates, the fund is also targeting municipalities with a specific and different pre-treatment trend. In the appendix, the first graph is showing the trend of groups organized by treatment date. Remarkably, there is only one group with lower-than-control-group average deforestation levels, which is also the one with the most similar trend, namely the 2016 group.

5 Results

Table 1 is showing results for the first simple Difference-in-Difference model, which starts with three variables; the year indicator, a treatment indicator and the interaction of both variables which, in this case, is called *did*. The dependent variable in this first specification is the cleared area in square kilometers. The interesting part of this simple model is that both the *Treated* and the *did* variables are highly significant, with large standard errors. Even more interesting, is the fact that the treatment effect, the *did* coefficient, is positive and relatively large. Receiving fund resources is correlated with higher deforestation levels, and we're back to the police presence and crime rate story. Additionally, when adding

control variables to the model[5], the previously insignificant coefficients (year and intercept) gain significance, and the treatment effect becomes smaller.

	Model 1 - Dependent variable is the cleared area
(Intercept)	2367.10
	(1494.36)
Treated	-11693.44^{***}
	(2238.07)
Year	-1.16
	(0.74)
did	5.81^{***}
	(1.11)
R^2	0.02
Adj. R^2	0.02
Num. obs.	1885

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table 3: DiD model 1

[5]Models with controls are in the appendix

Figure 3 shows the aggregate year means of control and treatment units. The figure does not show any treatment effect, because treated units did not get access to treatment in the same years. It shows, nevertheless, that the treated units have, on average, higher deforestation rates throughout the decade. The initial average forest loss (2008) of treated municipalities is more than twice as large as the initial one of non-treated municipalities, which goes in line with the large *Treated* coefficient, which shows the initial difference between treated and control units. It is also worth noting that 2018-2019 marks a radical change in environmental policies, which, on the graph, can be seen as the spike both in treated and control units deforestation rates. Treated Municipalities almost double their average deforestation rate in 2019, which also indicates these municipalities might be more vulnerable to rising illegal deforestation activities, thereby being targeted by the fund. Another important characteristic illustrated in figure 3, is that the control group deforestation means end up half of what they were in 2008 by 2020, while the treatment group deforestation means are substantially higher in 2020 relative to 2008. The trend followed by the control group confirms the negative *year* coefficient of most estimated models.

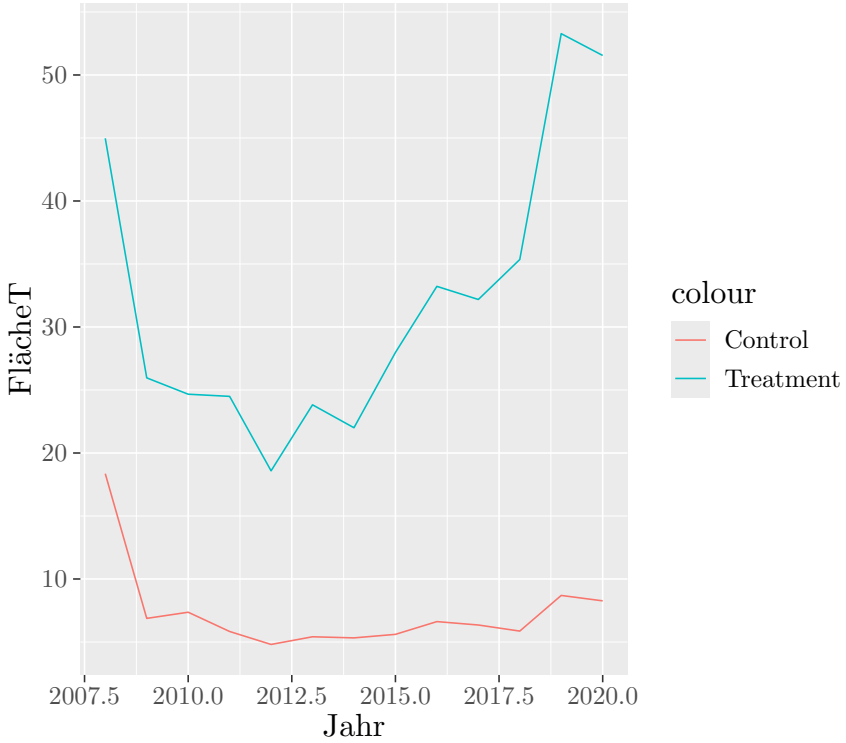


Figure 3: Treatment and Control means

Table 4 is showing the same model with the dependent variable being the cleared area relative to the municipality's total size. Apart from getting smaller, the coefficients gain significance, especially the *year* coefficient and the intercept, that were insignificant in the first version of the model. Although these results are confirming the ones of the previous version of the DiD model, when clustering standard errors at years, the first version of the Diff-in-Diff model remains similar in the significance of its coefficients. The second version however looses significance for all its coefficients, which suggests a violation of the Exogeneity/Independence assumption, leading us back to the endogeneity bias of the funds resource allocation previously discussed. The municipal trends are highly correlated to the introduction of treatment, as well as the outcome, leading to endogeneity.

	Model 2 - Dependent variable as share of total area
(Intercept)	37.41^{***}
	(9.03)
Treated	-93.57^{***}
	(13.52)
Year	-0.02^{***}
	(0.00)
did	0.05^{***}
	(0.01)
R^2	0.03
Adj. R^2	0.03
Num. obs.	1885

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table 4: DiD model 2

An alternative model leading to the opposite result of the two previous models uses the *diff* transformation for the dependent variable. The value for the dependent variable in year *t* becomes the difference of the cleared area in year *t* relative to year *t-1* (the value for year t = 2008 is NA as there is no previous year for a difference to be calculated). For the transformed data, it makes sense to use the *plm*[6] function, although we loose the intercept. This transformation has serious implications for the results, because rapid changes in deforestation rates, lead to very different values for the dependent variable. The results are therefore expected to contradict the ones of the two previous models.

Table 5 shows the coefficients for the same explanatory variables, and although there is little change in significance, the sign of the coefficients are now the opposites of the previous ones. The *treated* and *year* coefficients are now positive and, most importantly,

[6]Linear models for panel data estimated using the lm function on transformed data

the *did* treatment effect is negative, with the latter one being significant at the 5% level. The Fund could have an influence in stalling rising deforestation rates.

	Model 3 - Dependent variable is the difference in deforestation
Treated	2718.25*
	(1226.25)
Year	2.32***
	(0.45)
did	−1.35*
	(0.61)
R^2	0.03
Adj. R^2	−0.06
Num. obs.	1740

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table 5: DiD model 3

	Model 4 - Dependent variable is the difference in relative deforestation
Treated	44.54***
	(11.45)
Year	0.03***
	(0.00)
did	−0.02***
	(0.01)
R^2	0.04
Adj. R^2	−0.05
Num. obs.	1740

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table 6: DiD model 4

This counterfactual dependent variable is important, because rejects any of the previous conclusions that the Fund had a contribution to illegal activities, albeit it does not lead us closer to finding a causal effect. Using the *diff* transformation on the second specification for the dependent variable, i.e. cleared area relative to the municipality's size, leads to similar results except that the treatment indicator and treatment effect coefficients gain significance. It is also worth pointing out, that these two models using differences loose

significance when clustering standard errors unlike the previous ones [7].

Finally, Figure 4 is showing the staggered Difference-in-Difference deforestation rates over the years. The coefficient for year 0, the year of treatment is the omitted one, as all other coefficients are relative to this year. The strong initial decline is largely due to the overall downward trend seen during that initial period. The coefficient for the initial deforestation rate, *lead9* also indicates that most treated units had a very high initial deforestation rate, meaning that their inability to contain deforestation in times of increased overall deforestation is historical. The coefficient for *lead1*, one year before treatment, is also one of the highest, suggesting that most treated municipalities were experiencing an increasingly larger deforestation rate up until treatment, which confirms the endogeneity bias of the treatment introduction. The deforestation rates continue to rise after the year of treatment, confirming that the treatment is indeed correlated to higher deforestation rates, even if, they don't end up as high as before the treatment. One could argue that the funds response to increasing deforestation was to stall or retard the deforestation rates directly at municipalities, which wouldn't be unreasonable, since the lags never end up as high as the leads. Even so, it cannot be said that the Funds response had a significant impact on reducing deforestation.

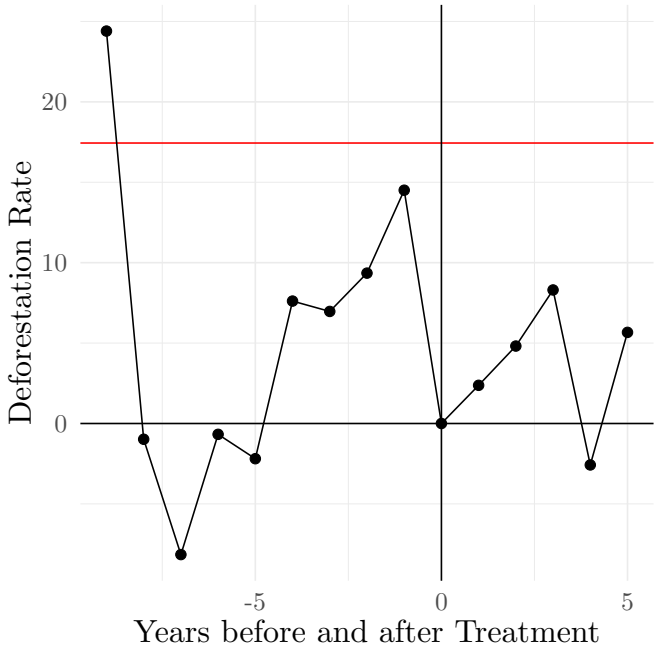


Figure 4: Staggered DiD - Leads and Lags

[7] Coefficient tables with clustered standard errors are shown in the appendix

A possible substitute for the Difference-in-Difference models is the Fixed Effects framework, and more specifically the *within* estimator, which uses the time-variation within each cross-section. The within estimator is calculated using year fixed effects, entity/municipal fixed effects as well as different transformations for the dependent and explanatory variables. The first specification is the same as the first DiD model, except that the main explanatory variable is the total amount of money that municipality i received in year t. Interacting this variable with individual years gives us a different treatment effect for each year. Moreover, there is an interesting finding, which is the fact that some year coefficients are highly significant. From 2011 to 2015, these coefficients are negative and vary in significance. The year coefficients of 2016 to 2018 remain negative but become insignificant, while the 2019 and 2020 coefficients become highly significant, large and, above all, positive. These coefficients are probably capturing the effects of the aforementioned omitted environmental policy variable over the years, with the last two of them reproducing a radical change in the overall approach of environmental policy. Both controls have a small coefficients, and curiously, the agriculture variable has a negative impact on the deforestation of a municipality, while the indigenous reservation variable has a positive, and highly significant coefficient. Using different variables in the fixed effects model yields similar results, with the individual year coefficients always being significant. The interactions between treatment and the individual years rarely become significant, except for the interaction with year 2012 which, in most model specifications, is negative. This year marked one of the lowest aggregate deforestation rates up to date, so the fact that the interaction is significant does not surprise.

The Fixed Effects model is written such as the following formula:

$$Y_{it} = \theta Fund_{it} + \gamma Year_t + \delta(Fund_{it} * Year_t) + \beta * (Fund_{it} * Municipality_i) + \epsilon \qquad (2)$$

	Model 1
Cashflowlags	−0.00
	(0.00)
factor(Jahr)2010	−0.31
	(1.14)
factor(Jahr)2011	−1.60
	(1.14)
factor(Jahr)2012	−4.12***
	(1.15)
factor(Jahr)2013	−2.37*
	(1.15)
factor(Jahr)2014	−3.58**
	(1.15)
factor(Jahr)2015	−2.70*
	(1.16)
factor(Jahr)2016	−0.66
	(1.16)
factor(Jahr)2017	−1.67
	(1.17)
factor(Jahr)2018	−1.88
	(1.17)
factor(Jahr)2019	4.10***
	(1.20)
factor(Jahr)2020	2.91*
	(1.19)
Reservations	0.00***
	(0.00)
agriculture	−0.00***
	(0.00)
R^2	0.11
Adj. R^2	0.02
Num. obs.	5964

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table 7: DiD model

6 Conclusion

It was found the municipal projects financed Amazon Fund had no significant impact on the forest loss of municipalities. The main finding of this seminar paper, is that municipalities with a specific profile were targeted and supported by the Fund. On average, these municipalities have had a higher experience with illegal deforestation, they are larger, and received Fund resources as their annual forest loss was on the rise, leading to a strong correlation between the Funds presence in a municipality and its forest loss. Nonetheless, further research could point to a conclusion that without the Fund, the forest loss in these municipalities would be even higher. The most likely way the Fund could work at a macro-level, would be if it had a significant effect on the forest change for the municipalities it is treating. The Fund could've slowed down the deforestation increment, leading these municipalities to have a stable annual deforestation loss, which could eventually be brought down with direct environmental activity coming from the government. Any further research, however, relies on solving the endogeneity problem of treatment allocation, in order to find the causal effect.

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A Appendix

A.1 Not included in the report

The measure of the dependent variable led to significantly different results. In a subject such as environment and forest loss that can be expected. The ideal way of constructing the dependent variable would be using a normalized measure constructed as the following:

$$Deforestation_{it} = (Area_{it} - \overline{Area_{it}}) * sd(Area_{it})^{-1} \tag{3}$$

This measure is used in many of references in this seminar and could've led to interesting results in the DiD framework.

A.2 *

Hiermit versichere ich, dass ich die vorliegende Arbeit selbstständig und ohne Benutzung anderer als der angegebenen Hilfsmittel angefertigt, noch nicht einer anderen Prüfungsbehörde vorgelegt und noch nicht veröffentlicht habe.

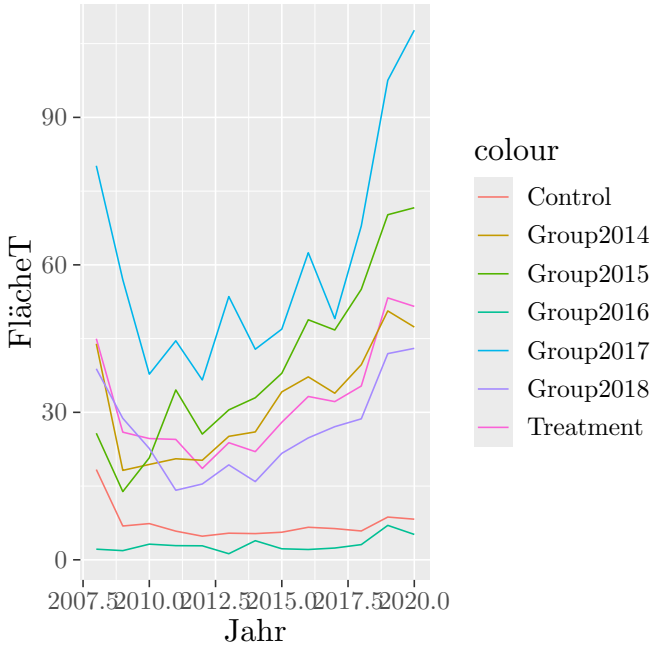


Figure 5: Treatment and Control means

München, 29/08/2021 Unterschrift:

	Model 1 - Clustered Standard Errors at years
(Intercept)	2367.10
	(1497.16)
Treated	-11693.44^{***}
	(2223.65)
Jahr	-1.16
	(0.74)
did	5.81^{***}
	(1.10)

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table 8: DiD model 1

	Model 1 - Clustered Standard Errors and controls
(Intercept)	3477.20^{**}
	(1284.27)
Treated	-9830.47^{***}
	(1773.96)
Jahr	-1.72^{**}
	(0.64)
did	4.88^{***}
	(0.88)
Reservations	0.00^{***}
	(0.00)
agriculture	0.00^{***}
	(0.00)

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table 9: DiD model 1

	Model 2 - Clustered Standard Errors at years
(Intercept)	37.41
	(1497.16)
Treated	−93.57
	(2223.65)
Jahr	−0.02
	(0.74)
did	0.05
	(1.10)

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table 10: DiD model 2

	Model 2 - Clustered Standard Errors and controls
(Intercept)	36.17
	(1284.27)
Treated	−93.98
	(1773.96)
Jahr	−0.02
	(0.64)
did	0.05
	(0.88)
Reservations	−0.00
	(0.00)
agriculture	0.00
	(0.00)

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table 11: DiD model 2

	Model 3 - Clustered Standard Errors at years
Treated	2718.25
	(1773.96)
Jahr	2.32***
	(0.64)
did	−1.35
	(0.88)

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table 12: DiD model 3

	Model 4 - Clustered Standard Errors at years
Treated	44.54
	(1773.96)
Jahr	0.03
	(0.64)
did	−0.02
	(0.88)

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table 13: DiD model 4